

29 Electron Orbits

Outcomes

- To recall more about
- the structure of matter
 - electron orbits
- To improve skills in
- understanding written information
 - understanding diagrams
 - drawing diagrams
 - recognising patterns
 - using models

Orbiting electrons

No-one has ever seen an electron. In fact, no one has ever seen an atom. The best technology we have can only give us information that we use to refine the model of the atom we have been developing.

A lot of the theories you learn about in Science will change as more information is found. Some of the ideas you learn about (and the structure of the atom is one of these) are very simplified ideas which do not give the true or complete picture. In cases like this it would be impossible to be able to understand the most recent theories because you don't have enough background knowledge. To get the background knowledge you need to work through older theories and learn why they have changed as science and technology have developed.

The ideas we have about how electrons are arranged in their orbits around atoms can be summarised, simply, in the following way:

1. Each atom has the same number of electrons as it has protons
2. Electrons are arranged in different orbits around the nucleus
3. The first electron orbit can only hold 2 electrons
4. The other orbits can hold more electrons according to the formula $2n^2$ where 'n' is the number of the orbit.

The table shows how many electrons can fit into the first four orbits of atoms:

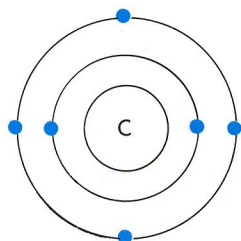
number of orbit	number of electrons orbit can hold ($2n^2$)
first orbit ($n=1$)	$2n^2 = 2$
2nd orbit ($n=2$)	$2n^2 = 8$
3rd orbit ($n=3$)	$2n^2 = 18$
4th orbit ($n=4$)	$2n^2 = 32$

5. The maximum number of electrons that can fit into the outermost orbit is 8

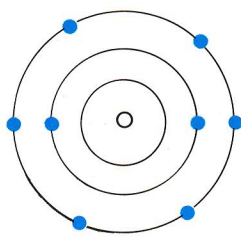
element	atomic number	number of electrons	arrangement of electrons in orbits
hydrogen	1	1	1 in first orbit
helium	2	2	2 in first (full)
lithium	3	3	2:1 (2 in 1st, 1 in second)
beryllium	4	4	2:2 (2 in 1st, 2 in 2nd)
boron	5	5	2:3 (2 in 1st, 3 in 2nd)
carbon	6	6	2:4 (2 in 1st, 4 in 2nd)
nitrogen	7	7	2:5
oxygen	8	8	2:6
fluorine	9	9	2:7
neon	10	10	2:8
sodium	11	11	2:8:1 (2 in 1, 8 in 2, 1 in 3rd)
magnesium	12	12	2:8:2
aluminium	13	13	2:8:3
silicon	14	14	2:8:4
phosphorous	15	15	2:8:5
sulfur	16	16	2:8:6
chlorine	17	17	2:8:7
argon	18	18	2:8:8
potassium	19	19	2:8:8:1
calcium	20	20	2:8:8:2

Fig 29.1 The Table above shows the arrangement of the electrons in their orbits for the first 20 elements in the periodic table.

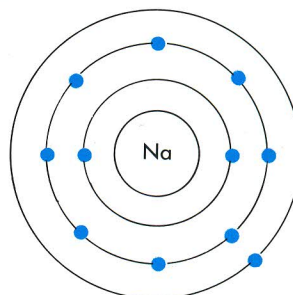
Electron shell diagrams can be drawn for each of these elements. These diagrams show the number of electrons in each shell or orbit.



Carbon has 2 electrons in its first orbit and 4 in its second orbit.



Oxygen has 2 electrons in its first orbit and 6 in its second.



Sodium has 2 electrons in its first orbit, 8 in its second orbit and 1 in its third and outer orbit.

Fig 29.2 The arrangement of electrons in their orbits as described in the table.

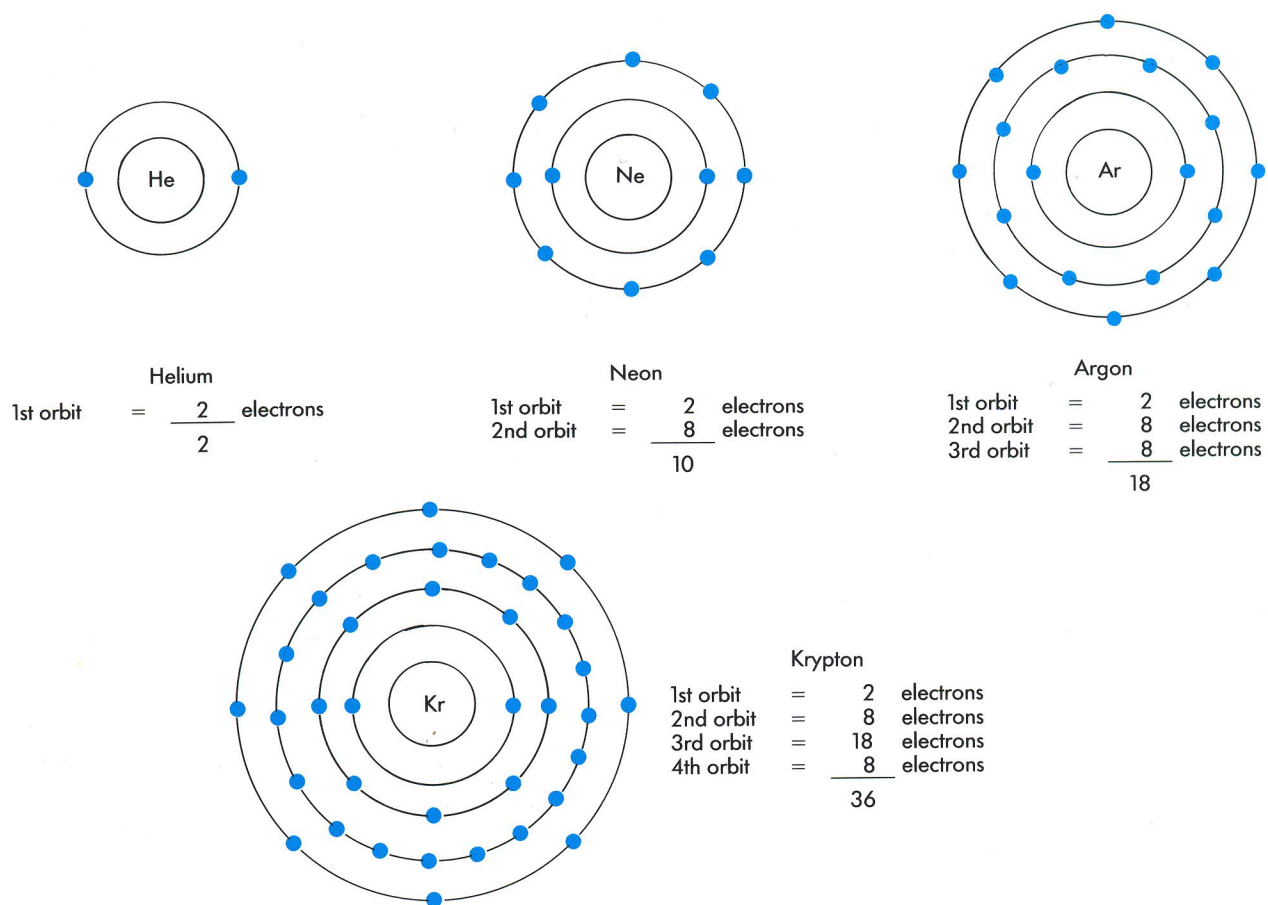


Fig 29.3 The inert (chemically inactive) gases, with 8 electrons in their outermost orbit are chemically inactive.

6. Atoms with a full outer orbit or with 8 electrons in their outermost orbit are very stable chemically and do not react with other atoms easily.

element	atomic number	number of electrons	arrangement of electrons in orbits
helium	2	2	2 in first orbit (full)
neon	10	10	2:8 (2 in 1st, 8 in second)
argon	18	18	2:8:8 (2 in 1, 8 in 2, 8 in 3)
krypton	36	36	2:8:18:8
xenon	54	54	2:8:18:18:8

The inert gases, except for helium, all have 8 electrons in their outermost orbit. They are chemically inactive. They are very stable.

7. Atoms with fewer than 8 electrons in their outermost shell (except for helium which has a full outer orbit) are chemically active.

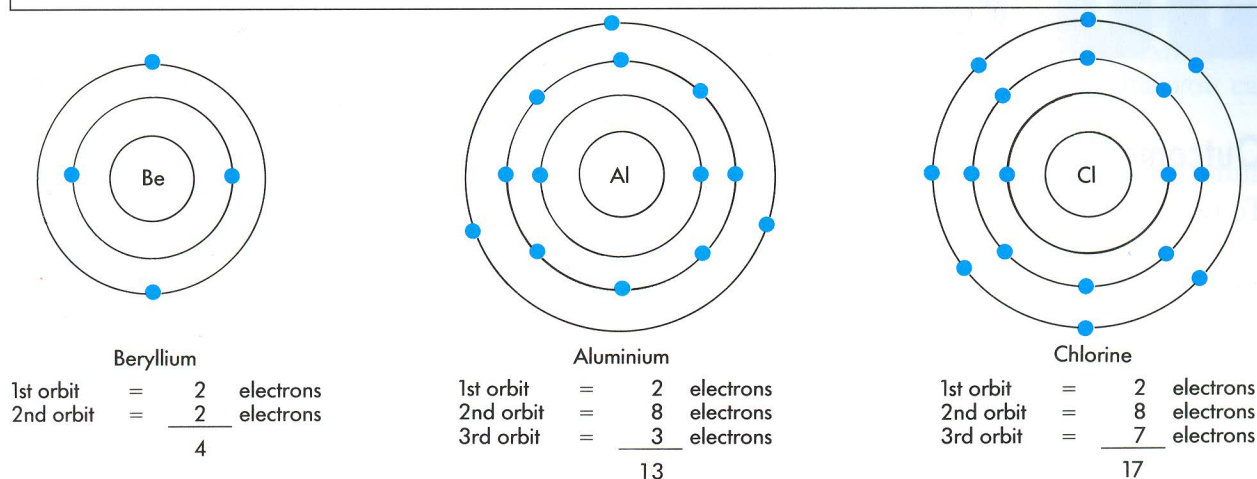


Fig 29.4 Elements whose atoms do not have 8 electrons in their outermost orbit (except neon, which has a full outer orbit) are chemically active.

Questions

- What is the maximum number of electrons that could fit into the 5th and 6th orbits of an atom?
- Element X has electrons arranged in its orbits like this:— 2:8:18:32:18:8. What can you say about element X?
- The chemical activity of an element is related to the number of electrons in its outermost orbit. Elements with 1 or 7 electrons in the outermost orbit are found to be more reactive than elements with 2 or 6 electrons in that orbit. In turn, these are more active than elements with 3 or 5, which are more active than elements with 4. The graph in Fig. 29.5 roughly summarises this idea. Copy the graph into your book and label its axes. Give the graph a title.
- Copy (or photocopy) the periodic table (figure 28.5) into your book. Add the information about electron arrangements from tables 29.2 and 29.3 to it. From this limited amount of information, what patterns can you see?
- Where, in general terms, do you think element X (Q2) would fit into the periodic table?
- Element Y has atomic number 24.
 - How many protons are in the nucleus of an atom of Y?
 - How many electrons does it have in orbit?
 - How many electrons does it have in its outermost orbit?
 - Use your periodic table to identify element Y.
 - Draw a diagram to show the nucleus and all the electron orbits for element Y. Show how many electrons are in each orbit.

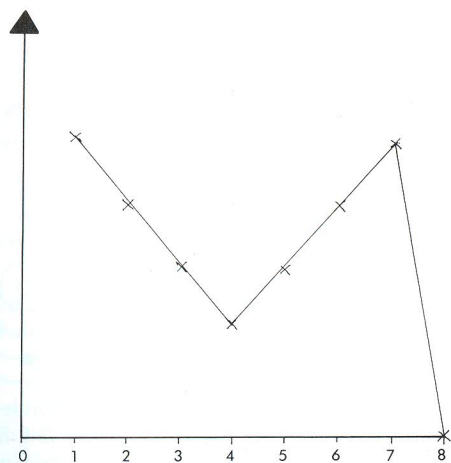


Fig 29.5